

CROTONE, ItalyWMO: **163500**Lat: **38.997N**Lon: **17.080E**Elev: **521**StdP: **14.42**Time Zone: **1.00 (EUC)**Period: **86-07**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	35.3	37.4	19.9	15.2	39.9	24.6	19.1	42.6	34.2	47.1	31.6	45.2	16.8	0	0.440

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	16.4	93.4	70.5	91.0	69.9	88.3	69.1	75.6	85.1	74.1	83.6	72.7	82.4	8.9	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
73.1	125.4	79.8	71.3	117.9	78.7	69.7	111.2	78.2	39.4	84.9	38.0	83.3	36.7	82.5	84.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 27.6	23.8	20.8	30.1	99.5	4.0	5.3	27.2	103.3	24.8	106.4	22.6	109.4	19.6	113.3		
(5)			28.0	79.0	2.2	2.4	26.4	80.7	25.1	82.1	23.8	83.5	22.2	85.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	62.1	49.0	48.8	51.9	56.6	64.0	71.8	78.6	79.0	72.5	65.4	56.3	50.4
	DBStd	11.90	4.45	4.95	5.35	3.61	5.17	5.26	4.31	4.13	4.29	4.34	5.06	5.12
	HDD50	250	71	71	42	2	0	0	0	0	0	0	9	55
	HDD65	2464	496	453	407	254	83	6	0	0	2	48	262	453
	CDD50	4672	40	38	100	199	435	654	888	900	676	476	199	67
	CDD65	1412	0	0	0	0	53	210	423	435	228	60	3	0
	CDH74	12151	0	0	0	1	294	1699	4293	4403	1337	123	1	0
CDH80	4514	0	0	0	0	34	509	1834	1803	321	13	0	0	
Wind	WSAvg	8.3	9.4	10.5	9.1	8.6	7.2	7.0	7.1	6.9	7.2	8.7	8.8	9.4
Precipitation	PrecAvg	36.5	4.5	3.5	3.2	2.3	1.7	1.0	0.9	1.0	2.8	5.6	5.0	5.0
	PrecMax	80.4	9.6	7.6	7.0	4.3	2.9	2.8	2.4	3.1	4.6	51.2	8.9	8.3
	PrecMin	24.5	2.0	1.1	0.2	0.1	0.6	0.1	0.1	0.1	1.0	0.7	2.2	0.6
	PrecStd	10.0	1.8	1.8	1.7	0.9	0.7	0.7	0.6	0.9	1.0	8.7	1.7	1.9
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	64.0	64.5	68.6	71.6	84.1	94.5	101.3	96.6	90.8	81.8	71.8	66.3
		MCWB	55.1	53.3	55.4	59.6	65.5	69.8	72.3	71.7	67.7	67.0	64.6	57.9
	2%	DB	60.4	60.9	65.9	68.3	80.2	88.5	93.7	93.3	86.2	76.9	68.8	62.8
		MCWB	54.0	52.7	55.6	58.1	64.7	67.6	70.8	70.5	67.7	66.3	62.7	57.6
	5%	DB	58.2	58.7	62.7	66.5	77.3	85.1	91.1	91.0	83.9	74.9	66.6	60.7
		MCWB	53.1	52.5	54.5	57.3	63.7	66.8	69.7	70.6	67.7	66.6	61.2	56.5
	10%	DB	56.8	57.0	60.5	64.6	74.4	82.5	88.8	88.3	80.9	72.8	64.7	58.5
		MCWB	52.3	51.8	54.0	56.6	62.3	66.1	69.0	69.9	67.6	65.4	59.8	54.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	57.6	56.8	59.4	62.8	69.7	73.4	77.7	78.2	74.5	72.1	68.1	60.7
		MCDB	61.1	60.4	64.1	68.1	77.3	87.3	87.2	85.6	80.8	75.8	69.9	63.3
	2%	WB	55.5	55.3	57.6	60.5	67.4	71.0	75.6	75.8	72.7	70.1	64.5	58.8
		MCDB	59.0	58.1	62.1	65.7	75.5	82.1	86.9	83.9	78.7	74.0	67.2	61.7
	5%	WB	54.1	53.8	56.3	58.9	65.8	69.6	74.1	74.4	71.2	68.5	62.2	57.2
		MCDB	57.6	57.3	60.5	64.3	73.8	80.0	85.2	83.1	77.9	72.5	65.6	60.1
	10%	WB	52.6	52.5	54.9	57.6	64.1	68.2	72.4	73.1	69.9	67.0	60.3	54.9
		MCDB	56.0	56.1	59.0	62.4	71.9	78.1	83.3	82.9	77.2	71.4	64.2	58.1
Mean Daily Temperature Range	5% DB	MDBR	11.4	11.7	12.8	14.0	15.8	17.7	17.1	16.4	14.7	12.3	11.9	11.5
		MCDBR	14.6	14.9	17.3	17.1	19.2	20.4	20.2	19.6	18.1	13.8	13.9	14.0
		MCWBR	10.9	10.7	11.6	11.0	9.4	9.1	9.4	9.7	9.5	9.1	10.4	10.7
	5% WB	MCDBR	12.5	12.6	14.3	14.7	17.2	18.2	17.7	16.9	14.8	12.4	11.8	12.4
		MCWBR	10.2	10.3	10.7	10.8	9.9	10.0	10.2	10.0	9.5	9.2	10.0	10.4
Clear-Sky Solar Irradiance	taub		0.345	0.369	0.408	0.447	0.441	0.423	0.421	0.431	0.430	0.404	0.375	0.351
	taud		2.439	2.351	2.240	2.144	2.218	2.313	2.333	2.305	2.298	2.342	2.384	2.426
	Ebn at Noon		260	268	268	264	268	272	272	266	258	252	246	248
	Edh at Noon		26	33	40	47	45	41	40	40	38	33	28	25
All-Sky Solar Radiation	RadAvg		653	902	1316	1721	2080	2338	2366	2099	1559	1098	728	580
	RadStd		61	84	113	120	94	120	87	92	90	92	57	64